

The Role of GPUs in Modern Data Science: Performance, Cost, and Computational Demand

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Introduction

- Heavy computational power is needed to handle large datasets and train complex models.
- GPUs are better suited for parallel processing than CPUs, making them very effective for training neural networks, image recognition, and large-scale data analysis [1].
- Such high-performance hardware can be expensive and energy-intensive, raising sustainability and equity issues in the field [2].
- This poster aims to show how computational demands in machine learning (ML) scale with data size, and how GPU performance, cost, and power characteristics affect modern data science workloads.

Computational Cost of Machine Learning Models

- To investigate computational cost of ML models as dataset size increases, three models were evaluated on synthetic data: Decision Tree, Random Forest, and K-Nearest Neighbours (KNN).
- Models were trained using 30 repeated timed runs for each dataset size.
- Time was recorded and averaged to provide a measure of computational cost.

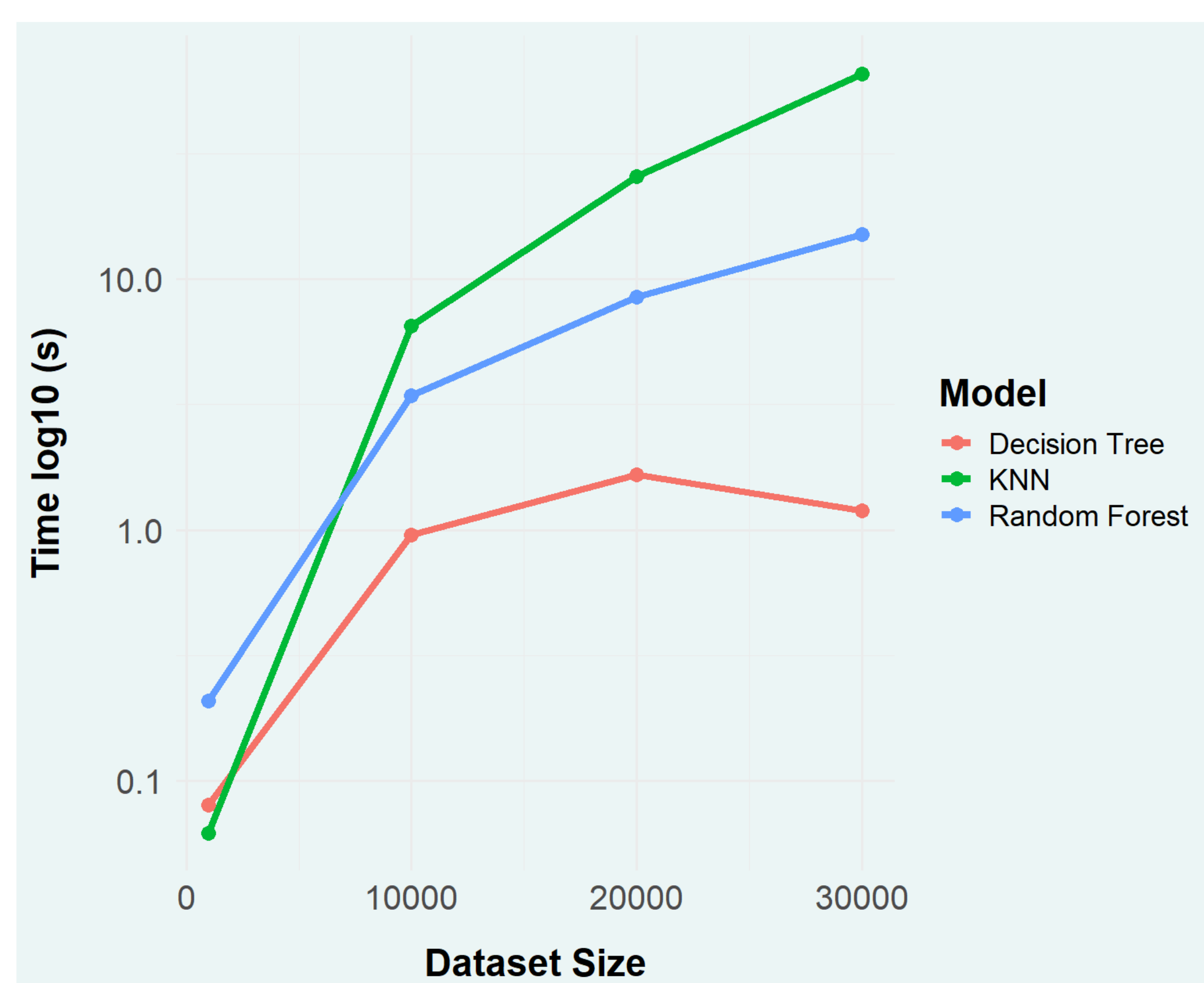


Figure 1: Logarithmic graph of training time for three machine learning models on datasets of increasing magnitude

- Computational cost generally increases with dataset size for all models, although they each scale differently.
- Minor fluctuations exist (e.g. Decision Tree) but general trend is upwards.
- As a result, more powerful hardware is required to cope with larger and more complex datasets.
- GPUs can provide the capability for parallel computations and allow for better performance when training larger, more complex models.

GPU Performance Over Time

- GPU benchmark scores from 2010–2022 were analysed to observe how computational capability has evolved over time [3].

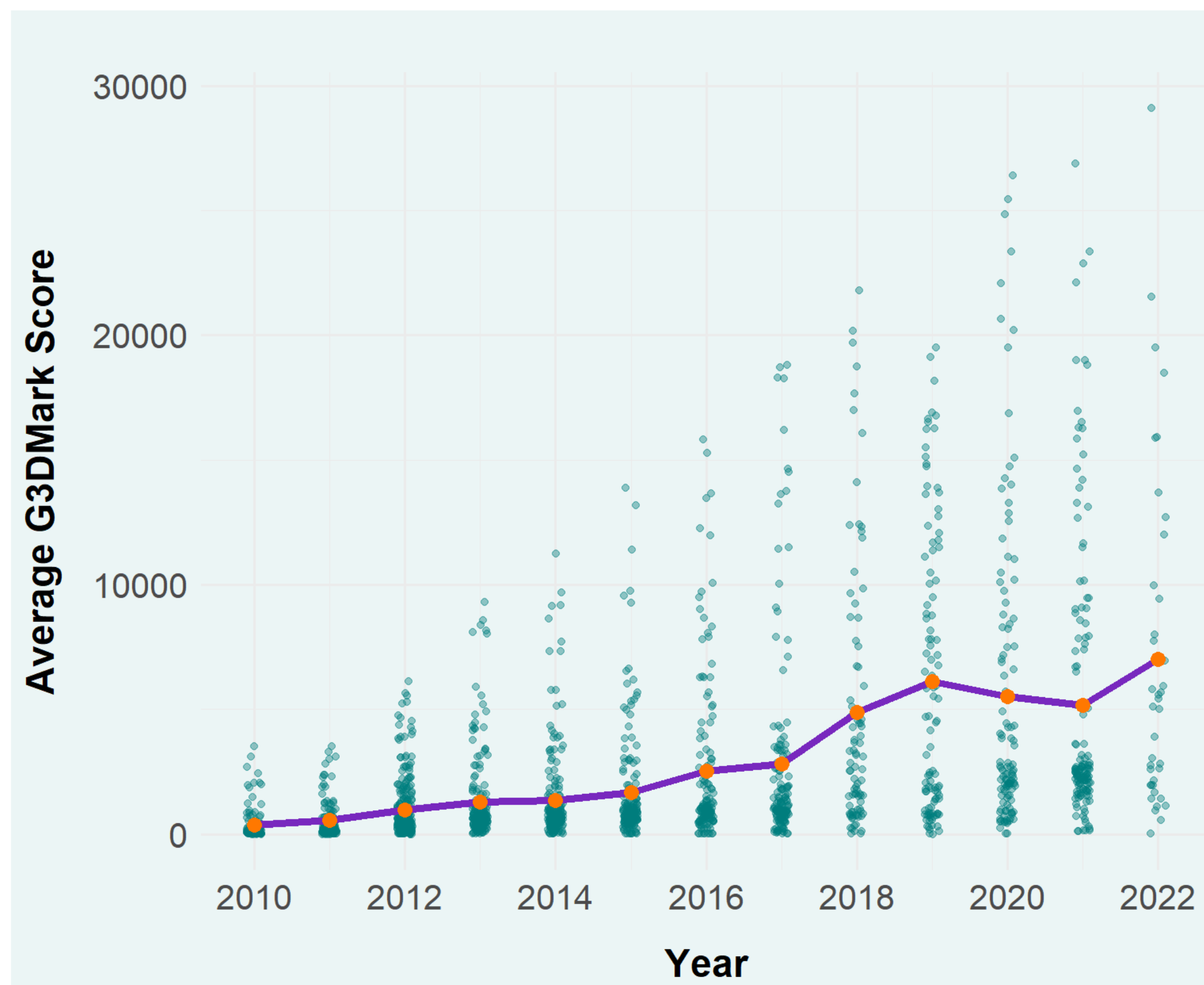


Figure 2: Evolution of GPU performance from 2010 - 2022 (Dots represent raw benchmark score data)

- GPU performance has steadily increased, with peak benchmark scores in 2022 being approx. 7 times higher than in 2010.
- Modern GPUs are far more capable at large-scale and complex modelling than in the past, allowing them to support the increasing computational demand of modern data science, and assisting in its development.

Price vs. Performance

- To evaluate how GPU performance scales with cost, benchmark scores were compared against market value [3].

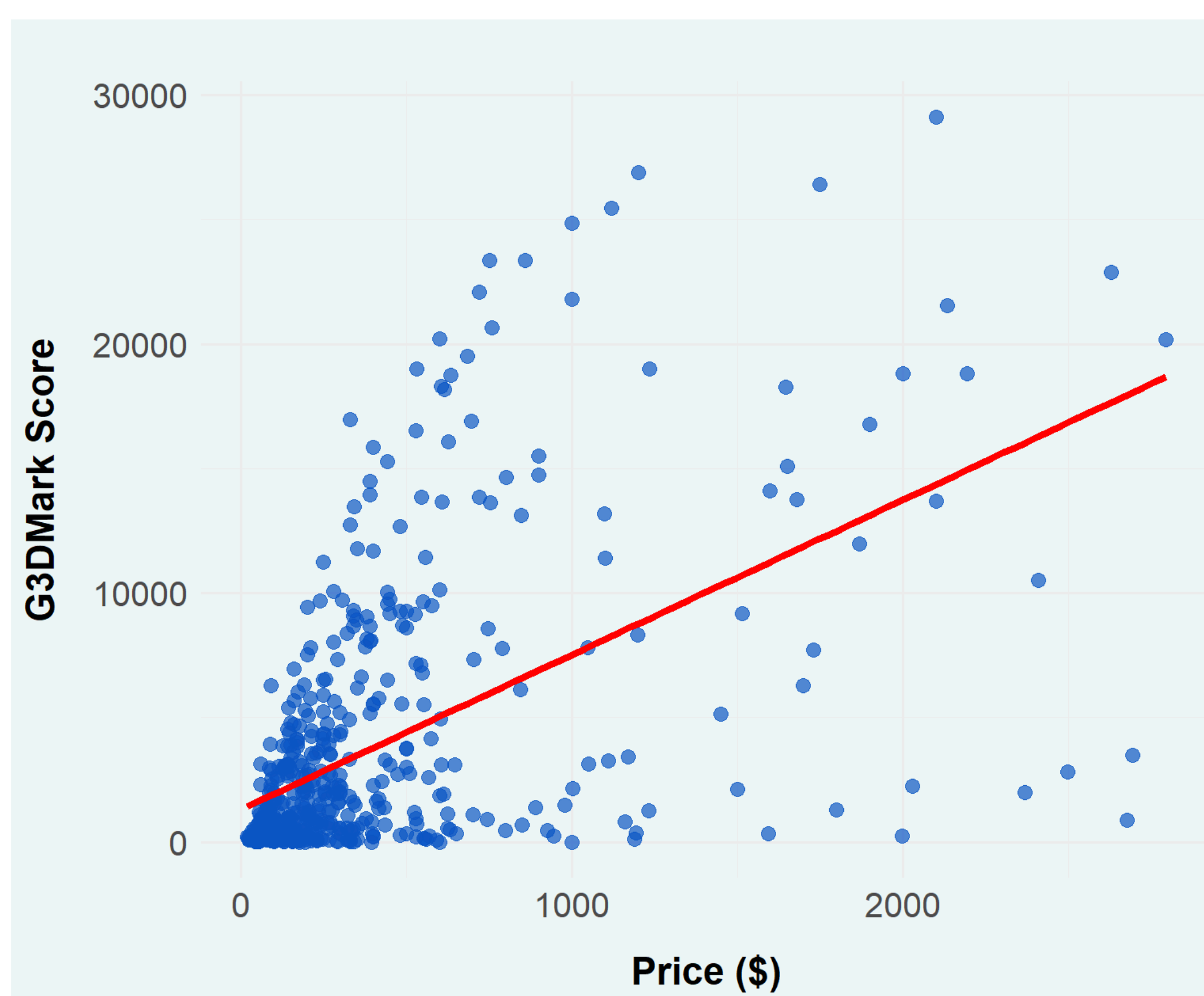


Figure 3: Relationship between GPU price and performance

- GPU performance generally increases with price.
- While the peak of GPU performance is found at the highest price points (≈ 2000 USD), excellent baseline capability is found at lower price levels as well (≈ 800-1000 USD), suggesting that cost does not solely determine performance.
- Efficient hardware selection can enable strong data science performance without requiring the most expensive GPUs.

Power vs. Performance

- To evaluate how GPU performance scales with energy usage, thermal design power (TDP) was compared against benchmark scores [3].

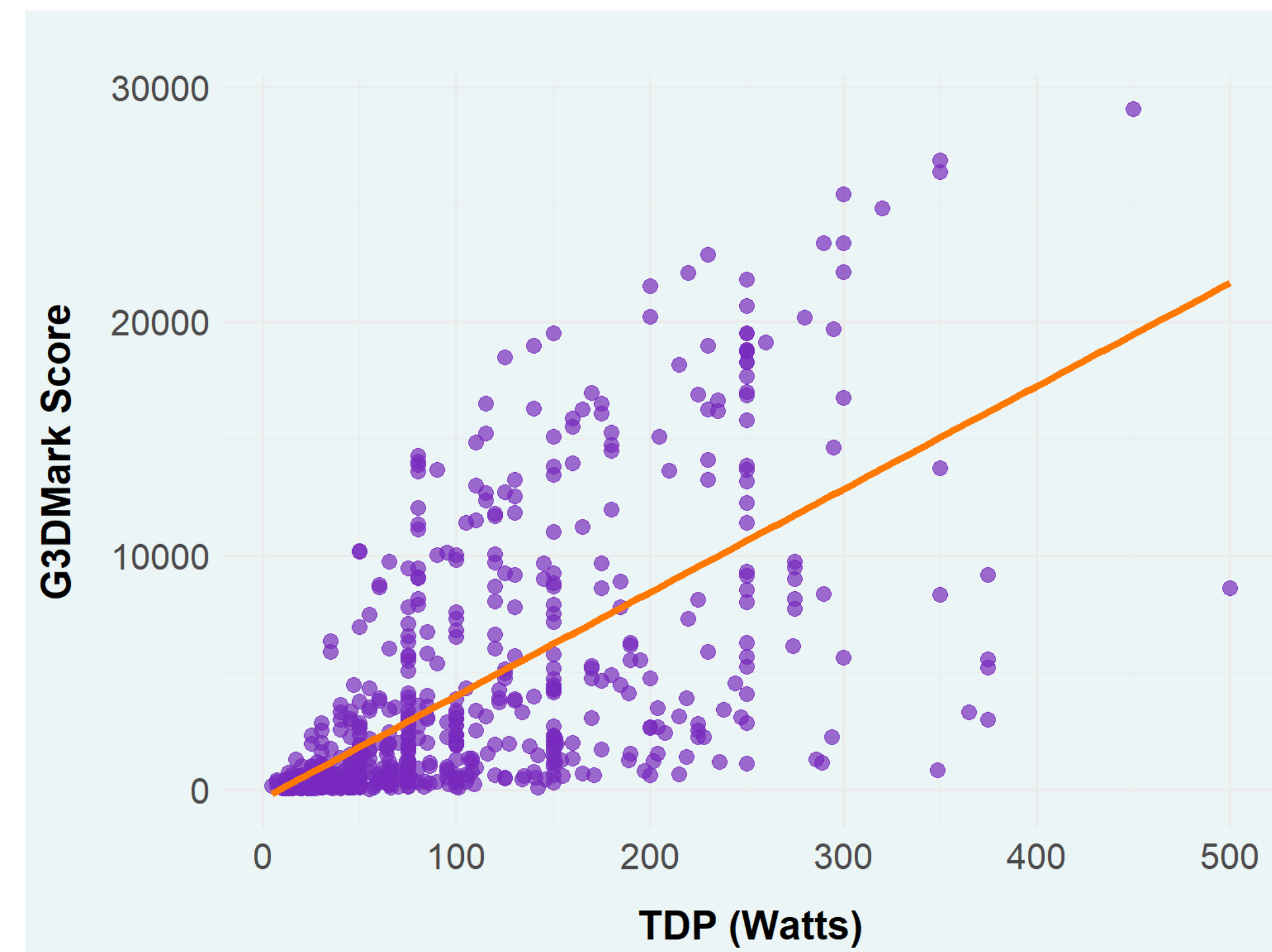


Figure 4: Relationship between GPU TDP and performance

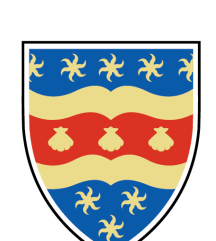
- GPU performance generally increases with power consumption. However, there is substantial variation in performance at similar power usage levels.
- While peak performance is found at higher power usage, high performance GPUs can be found at moderate TDP levels.
- Again, efficient hardware selection can allow for effective data science performance without requiring excessive power consumption, thereby reducing costs associated with such.

Conclusion

- GPUs support modern data science by providing the required processing capabilities needed for complex datasets and models.
- GPU performance has improved significantly over time, supporting the development and scalability of data science.
- While peak performance often comes with steep prices, high performance GPUs can be found at lower price points, reducing the issue of equity within the field.
- The same can be said for energy consumption, as efficient hardware selection can provide high performance with lower energy requirements, addressing sustainability concerns.

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